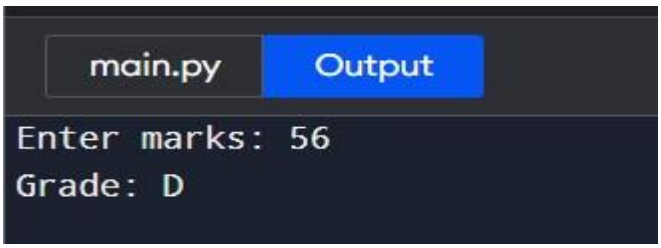
Experiment no:-4

Tile:- Implement if-else, nested if, and switch-case (using dictionaries) for decision-making scenarios.

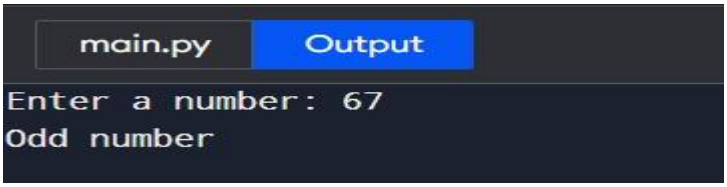
1.Write a program to check whether a given number is positive, negative, or zero using if-else marks

```
= int(input("Enter marks: ")) if marks >= 90:
   print("Grade: A")
elif marks >= 75:
print("Grade: B") elif
marks >= 60:
print("Grade: C") elif
marks >= 40:
print("Grade: D")
else:    print("Grade:
Fail")
```



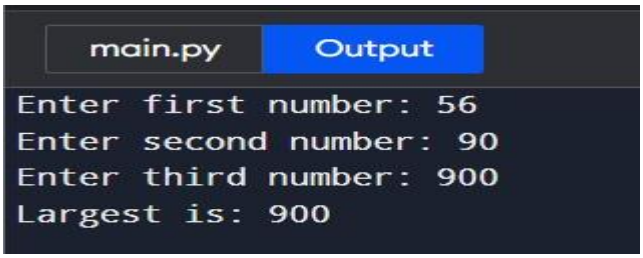
2.Write a program to check whether a number is even or odd using if-else.

```
num = int(input("Enter a number: ")) if
num % 2 == 0:
   print("Even number") else:
   print("Odd number")
```



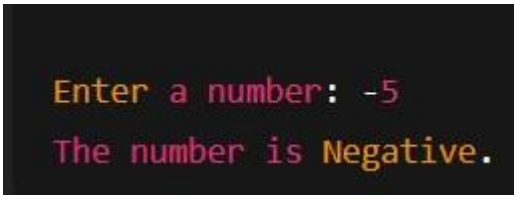
3. Write a program to find the largest of three numbers using nested if statements.

```
a = int(input("Enter first number: ")) b
= int(input("Enter second number: ")) c
= int(input("Enter third number: "))
if a > b:
if a > c:
    print("Largest is:", a)
else:
    print("Largest is:", c) else:
  if b > c:
    print("Largest is:", b)
else:     print("Largest
is:", c)
```



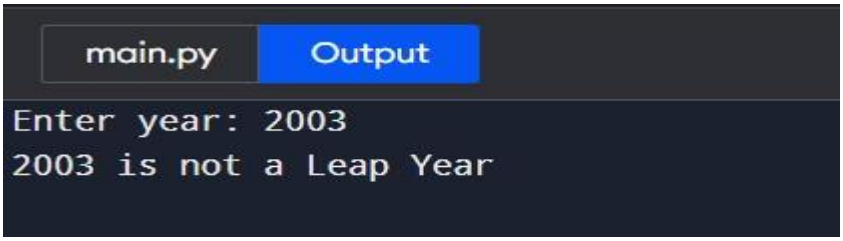
4. Write a program to check the number is positive or negative

```
num = float(input("Enter a number: ")) if num > 0:
  print("The number is Positive.") elif
num < 0:
  print("The number is Negative.")                    else:
  print("The number is Zero.")
```



5. Write a program to check whether a year is a leap year using if-else.

```
year = int(input("Enter year: ")) if (year % 400 == 0) or (year % 100 != 0 and year % 4 == 0):
  print(year, "is a Leap Year")
else:    print(year, "is not a Leap
Year")
```



6. Write a program to categorize age groups (child, teenager, adult, senior) using nested

if. age = int(input("Enter age: ")) if age < 13:

print("Child") else:

if age < 20:

print("Teenager")

elif age < 60:

print("Adult") else:

print("Senior Citizen")



7. Write a program to calculate electricity bill based on units consumed using if elif-else.

units = int(input("Enter units consumed: "))

if units <= 100:

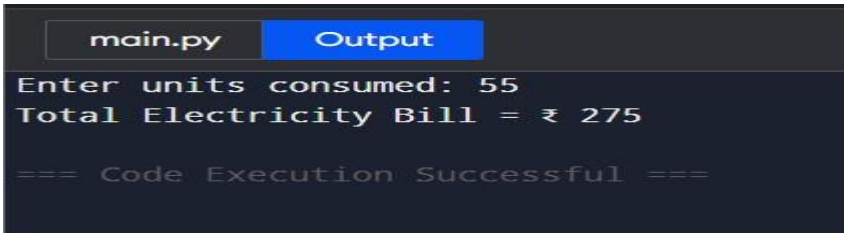
bill = units * 5 elif

units <= 200:

bill = 100*5 + (units-100)*7 else:

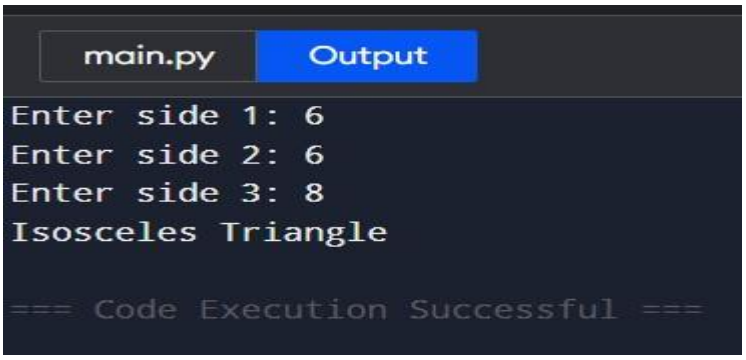
bill = 100*5 + 100*7 + (units-200)*10 print("Total

Electricity Bill = ₹", bill)



8. Write a program to determine the type of triangle (equilateral, isosceles, scalene) using nested if.

```
a = int(input("Enter side 1: ")) b
= int(input("Enter side 2: ")) c =
int(input("Enter side 3: ")) if a
== b == c:
   print("Equilateral Triangle") else:
   if a == b or b == c or a == c:
print("Isosceles Triangle")
else:        print("Scalene
Triangle")
```



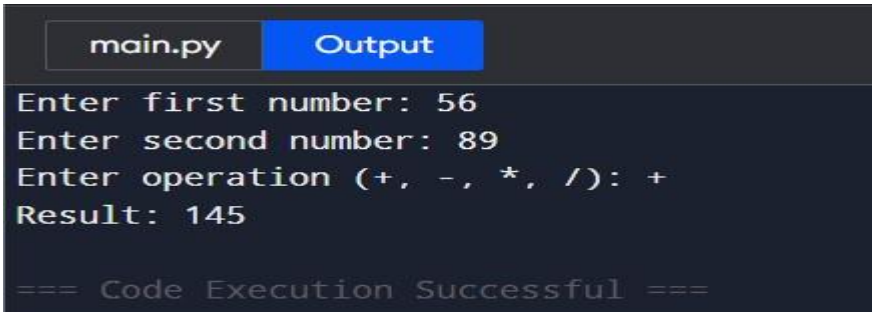
9. Write a program to print the day of the week based on a number input (1–7) using dictionarybased switch-case.



10. Write a program to perform arithmetic operations (+, -, *, /) based on user input using dictionary-based switch-case.

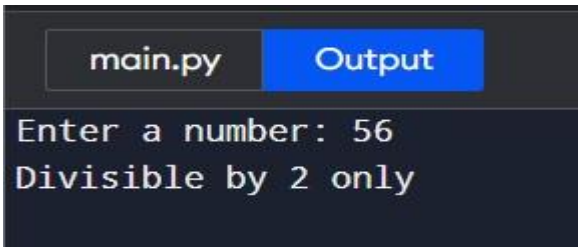
```
a = int(input("Enter first number: ")) b =
int(input("Enter second number: ")) op
= input("Enter operation (+, -, *, /): ")
operations = {
   '+': a + b,
   '-': a - b,
```

```
    '*': a * b,
    '/': a / b if b != 0 else "Cannot divide by zero"
}
print("Result:", operations.get(op, "Invalid Operation"))
```



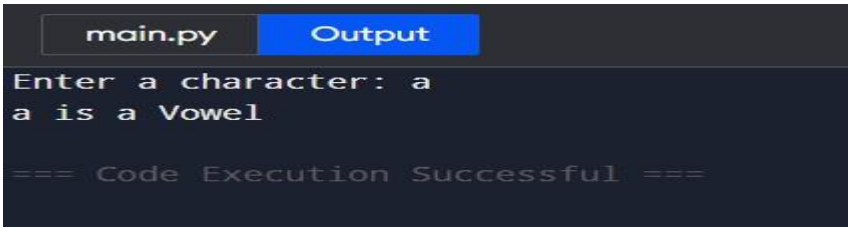
11. . Write a program to check whether a number is divisible by 2, 3, and 5 using nested if.

```
num = int(input("Enter a number: "))
if num % 2 == 0:     if num % 3 == 0:
if num % 5 == 0:
          print("Divisible by 2, 3, and 5")
       else:
          print("Divisible by 2 and 3 only")
   else:
       print("Divisible by 2 only") else:
print("Not divisible by 2, 3, and 5")
```



12. Write a program to check if a character is a vowel or consonant using if-else.

```
ch = input("Enter a character: ").lower()
if ch in 'aeiou':
   print(ch, "is a Vowel")
else:     print(ch, "is a
Consonant")
```



13. Write a program to map months (1–12) to their names using dictionary-based switch-case.



14. Write a program to implement a simple menu (1. Add, 2. Subtract, 3. Multiply, 4. Divide) using dictionary-based switch-case.

```
a = int(input("Enter first number: ")) b

= int(input("Enter second number: "))

menu = {

   1: a + b,

   2: a - b,

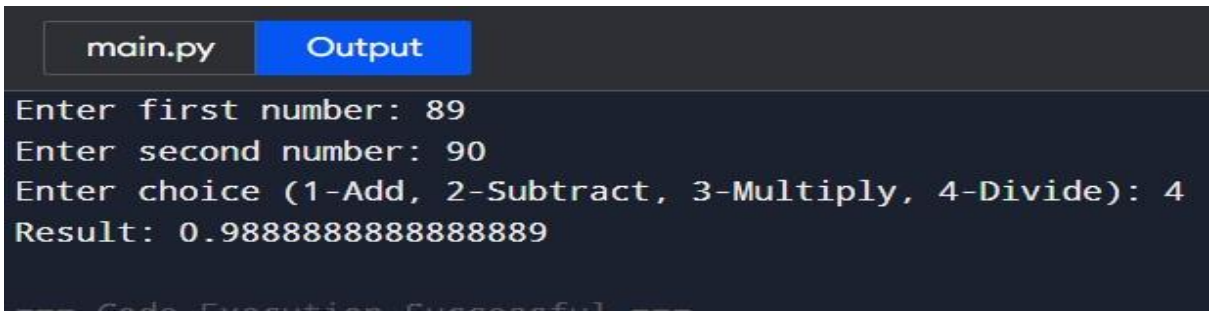
   3: a * b,

   4: a / b if b != 0 else "Cannot divide by zero"

}


choice = int(input("Enter choice (1-Add, 2-Subtract, 3-Multiply, 4-Divide): "))

print("Result:", menu.get(choice, "Invalid Choice"))
```



15. Write a program to determine if a student is eligible for a scholarship based on multiple criteria (marks, attendance) using nested if. marks = float(input("Enter marks: ")) attendance = float(input("Enter attendance percentage: "))

```
if marks >= 85:
    if attendance >= 75:
        print("Eligible for Scholarship")
    else:        print("Not eligible - Attendance
too low") else:     print("Not eligible - Marks
too low")
```

